October 23, 2026

To: Paul Taylor, Director, 311 & Insights, People, Innovation & Collaboration Services
To: Liz Ormsby, City Auditor, City Auditor's Office
Cc: Mayor Jeromy Farkas; Councillor Jennifer Wyness, Chair, Audit Committee; Councillors Andrew Yule (Ward 3), Raj Dhaliwal (Ward 5), Myke Atkinson (Ward 7) and Harrison M. Clark (Ward 9)
The City of Calgary, P.O. Box 2100, Station M, Calgary, AB T2P 2M5

Re: Who Calgary's 311 hears: an independent analysis of 311 reporting equity, with recommendations

Dear Mr. Taylor and Ms. Ormsby,

I am a Calgary resident and a CPA. Using only the City's open data, I analysed 809,528 non-duplicate 311 requests made between October 2024 and September 2026 across 195 communities. I am writing to share the results with you directly, because they point to an opportunity to make 311 work equally well for every Calgarian.

The good news is that once a problem is reported, the City responds just as quickly in every income and language group. The concern is upstream: communities where many residents do not speak English report far fewer infrastructure problems per resident. The raw gap is 39%, and after accounting for income, renters, community age and distance from downtown it is still about 23% (95% range 9% to 33%). In a system that fixes what people report, problems that are not reported can go unfixed.

This report sets out the three findings, where the gap is concentrated, and twelve practical recommendations, including a reporting-incentive pilot that could be tested in a few wards at low cost. The full methodology, every data file and the analysis code are public so that City staff can check and reproduce every figure.

The findings, an interactive community map and all downloads are at **hassanaliarshad110.github.io/calgary-311**.

I would welcome the chance to walk your teams through the analysis, and I would be grateful for any corrections or context the City can offer.

Sincerely,

Hassan Ali, CPA (ICPAP)
Calgary, Alberta
hasssanaliarshad@gmail.com

Who Calgary's 311 Hears

Equity in 311 reporting: findings and recommendations for The City of Calgary

Independent analysis of City of Calgary open data · 311 requests October 1, 2024 to September 30, 2026 · 195 communities, 1,279,690 residents · hassanaliarshad110.github.io/calgary-311

In short. 311 responds equally fast everywhere, but it hears much less from some communities. Where more residents have no knowledge of English, residents report 39% fewer infrastructure problems per person than in communities with the fewest non-English speakers. About 23% of the gap remains after comparing like with like. The likely cost is uneven maintenance: the City can only fix what it hears about.

Three findings

	Finding	Key figure
1	Communities where more residents don't speak English report far fewer infrastructure problems (roads, water and sewer, parks and trees).	90.0 vs 54.6 reports per 1,000 residents a year; about 23% fewer after adjustment
2	Once a problem is reported, the City closes requests equally fast in every income and language group.	Within 1.5% of the citywide pace at 7 days and 0.6% at 30 days
3	A smaller share of requests come through the 311 app in lower-income and non-English-speaking communities.	App share 18.6% where most residents don't speak English vs 23.4% where fewest don't

Top recommendations

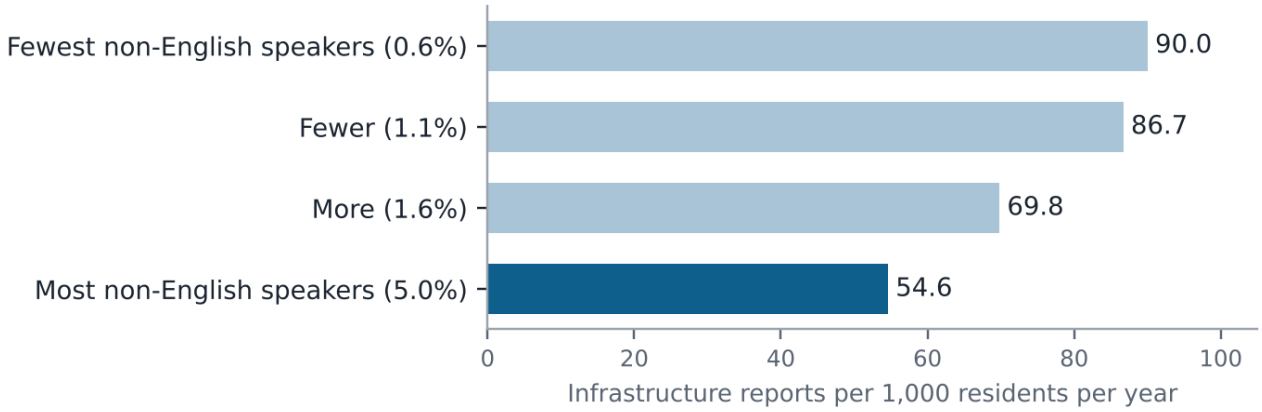
- **Measure who 311 hears.** Add reports per 1,000 residents by community to 311's performance reporting, next to response time.
- **Remove the language barrier.** Launch the multilingual app first in the communities with the largest gaps, and add photo-first reporting that needs no typing.
- **Reward reporting.** Pilot "Fix-It Points": verified, non-duplicate reports earn points redeemable for City recreation passes or transit tickets.
- **Work through trusted messengers.** Fund multilingual 311 ambassadors with settlement agencies and community associations, and a community challenge with small grants.
- **Don't wait for reports.** Send proactive inspections, including camera-based road defect detection, first to under-reporting communities.

Why this matters

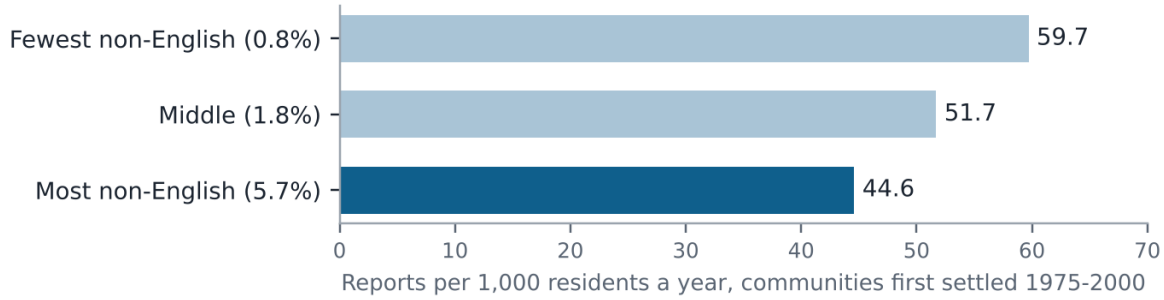
Much of the City's day-to-day maintenance responds to what residents report through 311: potholes, broken streetlights, damaged sidewalks, sewer backups, fallen trees. That is efficient, but it assumes every community reports at the same rate. If some communities report less because of language, time, trust or familiarity with the system, their problems are less likely to reach a work queue, even though the City responds fairly to the reports it receives.

Finding 1: some communities report far fewer problems

Communities were split into four equal groups by the share of residents with no knowledge of English (2021 Census). The quarter with the most non-English speakers reports far fewer infrastructure problems per resident.



Some of that raw gap reflects other differences, such as older neighbourhoods having older infrastructure. Among the 63 communities first settled between 1975 and 2000, the gap remains:

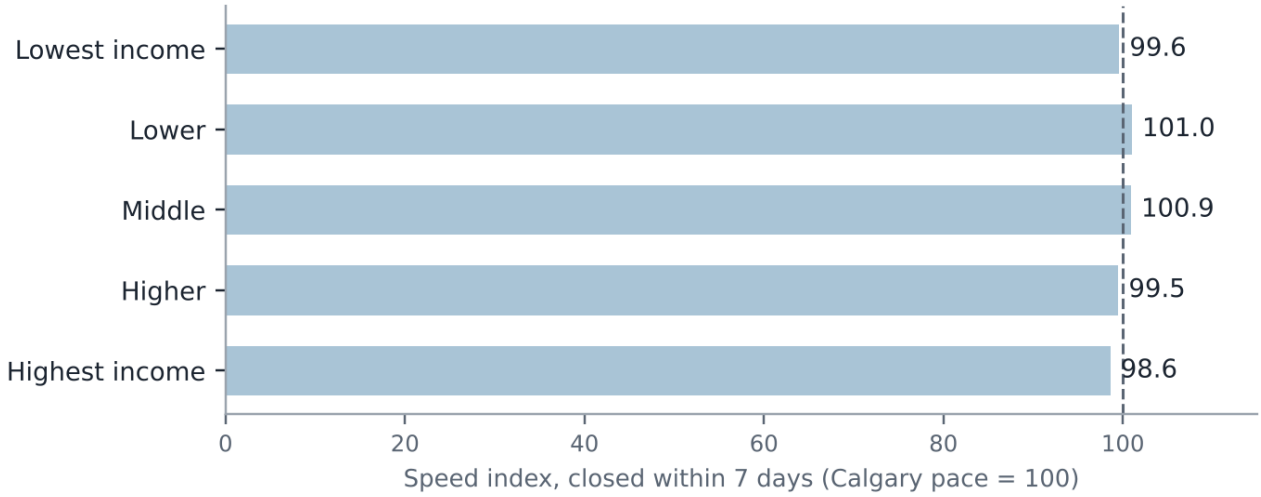


A regression that accounts for income, renters, community age and distance from downtown gives about 23% fewer reports for a community where 5% of residents do not speak English compared with one where 1% don't (95% range 9% to 33%). The result held in all 14 versions of the model tested: outside northeast Calgary, for roads, water and parks separately, in each year, and at area level.

Caution: language, visible-minority status and recent immigration go together across Calgary communities and cannot be separated with these data. The analysis shows who 311 hears less from, not exactly why.

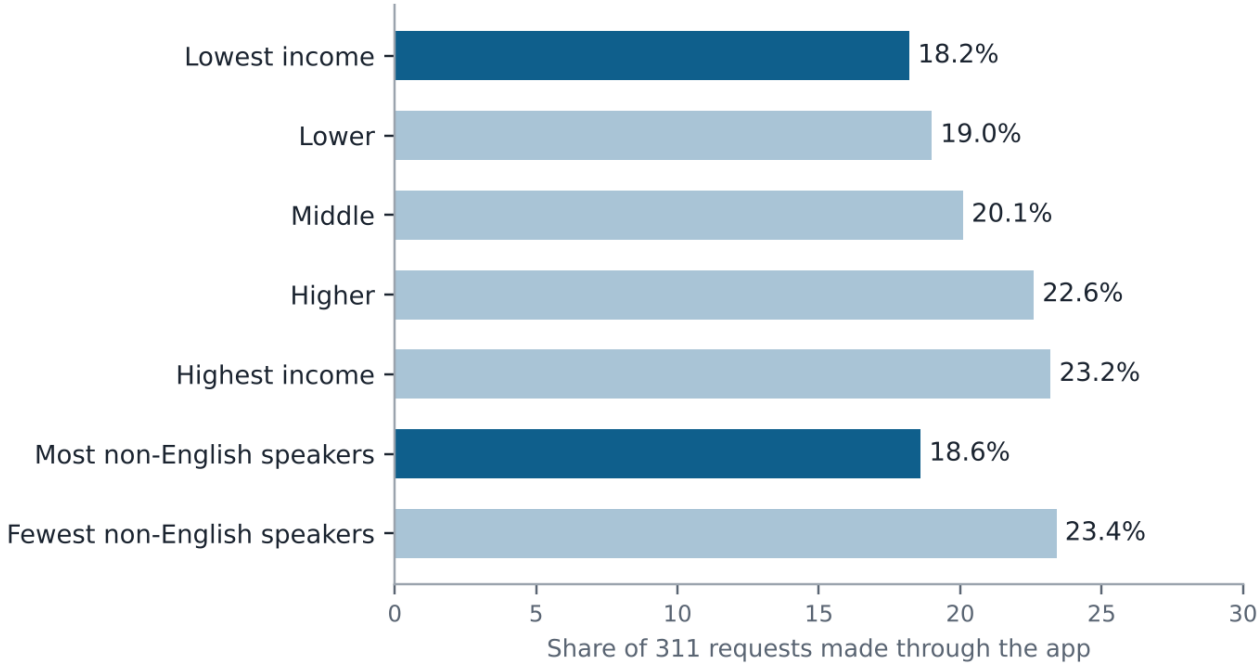
Finding 2: once reported, response is equally fast

Each community's requests were compared with the citywide speed for the same request type in the same year. A score of 100 means exactly the citywide pace.



This is a real strength of the 311 system and should be recognised. It also means the gap in Finding 1 is not a response problem; it happens before a request exists.

Finding 3: lower app use where barriers are highest



The City plans to add more languages to the 311 app in 2026. These are the communities where that change can matter most.

Where the gap is concentrated

The interactive map at the study website shows every community. The table lists the largest communities that combine a high share of non-English speakers (3% or more) with reporting at least 25% below what is typical for similar communities.

Community	Ward	Residents (2021)	No knowledge of English	Reports vs similar communities
Panorama Hills	3	25,535	7.0%	-41%
Taradale	5	17,630	9.1%	-34%
Martindale	5	14,540	9.5%	-27%
Hamptons	2	7,360	4.0%	-29%
Forest Heights	9	5,985	6.5%	-35%
Downtown West End	7	2,825	5.9%	-38%
Chinatown	7	2,250	5.9%	-42%
Greenview	4	2,100	6.4%	-40%

"Reports vs similar communities" compares a community's infrastructure reports per resident with what is typical for communities of the same income, renter share, age and distance from downtown. Language characteristics are measured for areas that group several neighbouring communities.

Recommendations

Grouped by what they change. Most can be piloted within existing budgets; the incentive and ambassador ideas are best tested in two or three wards first and measured against similar communities that do not get them.

#	Recommendation	Lead	Effort
	A. Measure who 311 hears		
1	Publish reports per 1,000 residents by community and ward as a 311 equity measure, alongside response time, in quarterly performance reporting.	311 & Insights	Low
2	Refresh this reporting-gap analysis every year with the City's full internal data, and share it with ward offices.	311 & Insights	Low
	B. Make reporting easier		
3	Launch the multilingual 311 app first in high-gap communities, with in-language promotion.	311 & Insights; Marketing & Communications	Medium
4	Add photo-first reporting: snap a photo, drop a pin, choose an icon. No typing or English needed.	311 & Insights	Medium
5	Accept reports by text message or WhatsApp in the most-spoken languages of high-gap communities, routed into the same 311 queue.	311 & Insights	Medium
6	Close the loop: tell every reporter when the problem is fixed, with a photo where possible. Seeing results builds trust and repeat reporting.	Service departments	Low

#	Recommendation	Lead	Effort
	C. Reward and encourage reporting		
7	**Fix-It Points:** verified, non-duplicate infrastructure reports earn points redeemable for City-owned rewards such as leisure centre drop-in passes or transit day passes. Cap points per person per month and award them only after City verification, so the program cannot be gamed.	311 & Insights; Recreation	Medium
8	**Community Care Challenge:** community associations in high-gap communities earn small grants for growth in verified reports per resident, with a public leaderboard by ward.	Community Services	Low to medium
9	**311 Community Ambassadors:** train multilingual residents through settlement agencies, faith centres and community associations to help neighbours report and to lead seasonal "walk and report" audits with ward offices.	Community Services; ward offices	Medium
10	Reach young people: school and youth-group "map your block" projects that report real issues, recognised through volunteer certificates.	Community Services	Low
	D. Don't rely on reports alone		
11	Schedule proactive inspections, including camera-based road defect detection the City has already tested, first in under-reporting communities.	Mobility; Water; Parks	Medium
12	Measure "fixed", not just "closed": track how often a closed request is followed by a new one for the same issue, by community.	311 & Insights	Low

Incentives carry risks: duplicate or trivial reports, and rewards flowing to residents who already report. Verification before points are awarded, monthly caps, and a pilot limited to high-gap communities address most of these. The City's own data will show quickly whether verified reports per resident rise.

A 12-month pilot the City could run

Element	Proposal
Where	Two or three wards with large reporting gaps (for example Wards 3, 5 and 9), with similar communities elsewhere as a comparison group.
What	Multilingual and photo-first reporting promoted locally, Fix-It Points, a Community Care Challenge, and trained ambassadors.
Measures	Verified infrastructure reports per 1,000 residents; share of new reporters; duplicate rate; repair backlog found by proactive inspection; resident awareness of 311.
Success test	Reporting in pilot communities moves at least halfway toward similar communities, with no rise in the duplicate rate.
Cost control	Rewards limited to City-owned passes with a fixed annual budget; ambassadors through existing settlement-agency partnerships.

How the analysis was done

- **Data:** City of Calgary open data: 311 Service Requests (every non-duplicate request made October 1, 2024 to September 30, 2026), 2021 Census by community, Calgary Equity Index, property assessments, building permits and historical community populations.

- **Scope:** 195 communities with 500 or more residents in 2021 (1,279,690 residents). Reporting rates use 30 common resident-reported request types; infrastructure means 21 types covering roads, water and sewer, and parks and trees.
- **Speed:** each request was compared with the citywide share closed within 7, 30 and 90 days for the same request type and year.
- **Estimate:** population-weighted regression on the log of reports per resident, controlling for income, renters, community age and distance from downtown; 14 model versions; a 1,000-draw bootstrap resampling whole areas for the 95% range.
- **Checks:** every figure was independently recomputed from the raw extracts, and the analysis code reproduces every result from the published data package.
- **Limits:** the analysis shows an association, not its cause. Fewer reports could partly reflect fewer problems. Income and language are measured for areas that group several communities, and population is from the 2021 Census.

The full methodology (with every model result), the community map and all data and code are at **hassanaliarshad110.github.io/calgary-311**.

This is an independent study. It is not affiliated with The City of Calgary, and it makes no allegation against any person, department or contractor.